PAPER_119: UQFF General Equation Systems – Compressed, Resonant, Buoyancy, Superconductive, Triadic, Quadratic, and Master Buoyancy with Full Variable Equations

Session: 0

Title: UQFF General Equation Systems – Compressed, Resonant, Buoyancy, Superconductive, Triadic, Quadratic, and Master Buoyancy with Full Variable Equations

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Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57, κ_i = 0.61)
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Index Slot: 1.16 UQFF Equation Systems Reference
Preceding Paper: PAPER_064 (4 simplified modes, Batch 23)

Abstract

The Unified Quantum Field Superconductive Framework (UQFF) operates in seven distinct equation systems, each derived from the master unified field F_U but applied at different scales and computational modes. This paper provides a complete variable-equation reference for all seven systems: (1) UQFF Compressed the compact unified form; (2) UQFF Resonant oscillatory/resonant terms; (3) UQFF Buoyancy the Ub_i opposition system; (4) UQFF Superconductive [SCm] reactivity coupling; (5) UQFF Triadic triadic master equations for plasma/turbulence systems; (6) UQFF Quadratic quadratic field approximations; and (7) UQFF Master Buoyancy extended Ub_i with Mayan-aligned time scaling. All variable equations are listed with their definitions, dimensions, and physical origins. This reference supersedes and extends PAPER_064 (four simplified modes) with the full-complexity forms extracted from the 393-page verification corpus.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^{-4} day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Master Unified Field Equation F_U

Before defining the seven systems, the master equation from which all are derived:

$$F_U = \sum_{i=1}^{4}\left[k_i U_{gi} - \beta_i U_{gi}\cdot\frac{\omega_g M_{bh}}{d_g}\cdot E_{react}\right] + \sum_j\left[\frac{\mu_j}{r_j}\left(1-e^{-\gamma t\cos(\pi t_n)}\right)\hat{\phi}_j\right] + \left(g_{\mu\nu}+\eta T_s^{\mu\nu}\right) - \sum_i [\delta_i$$

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2. System 1: UQFF Compressed (Compact Unified Form)

Long Form:

$$F_{U,\text{Compressed}} = \sum_{i=1}^{4}\left[k_i U_{gi} - \beta_i U_{gi}\frac{\omega_g M_{bh}}{d_g}E_{react}\right] + \sum_j \frac{\mu_j}{r_j}(1-e^{-\gamma t\cos(\pi t_n)})\hat{\phi}_j + (g_{\mu\nu}+\eta T_s^{\mu\nu}) - \sum_i \delta_i$$

Variable Equations:

Symbol	Value / Equation	Units	Description
F_U	Full sum of components	J/m	Unified energy density
i	14	–	Gravity range index
k_i	$k_1 = 1.5,\ k_2 = 1.2,\ k_3 = 1.8,\ k_4 = 1.0$	–	Coupling constants
U_{g1}	$k_1 \mu_s (M_s/r) e^{-\alpha t} \cos(\pi t_n)(1 + \beta_{def})$	N/m	Dipole gravity
U_{g2}	$k_2(\lambda_{vac,[UA]} + \lambda_{vac,[SCm]})(M_s/r^2)\, S(r - R_b)(1 + \delta_{sw} v_{sw}) H_{SCm} E_{react}$	N/m	Bubble gravity
U_{g3}	$k_3 \sum_j B_j \cos(\omega_s t)\, P_{core}\, E_{react}$	N/m	Disk strings gravity
U_{g4}	$k_4 \lambda_{vac,[SCm]} (M_{bh}/d_g) e^{-\alpha t} \cos(\pi t_n)(1 + f_{feedback})$	N/m	BH-star interaction
β_i	0.61 (uniform $i=1$4)	–	Buoyancy coupling
ω_g	7.3×10^{-16} rad/s	rad/s	Galactic spin ($v = 220$ km/s, $r = 8$ kpc)
M_{bh}	8.15×10^{36} kg	kg	BH mass (4.1×106 M?, Sgr A*)
d_g	2.55×10^{20} m	m	Galactic distance (27,000 ly)
E_{react}	$10^{46} e^{-0.0005t} = \rho_{vac,[SCm]} v_{SCm}^2 / \rho_{vac,A} \cdot e^{-\kappa t}$	W/m	[SCm] reactor efficiency
μ_j	$(10^3 + 0.4 \sin(\omega_c t)) \times 3.38 \times 10^{20}$	Tm	Magnetic moment of string j
r_j	1.496×10^{13} m	m	String distance (100 AU)
γ	0.00005 day^{-1}	day^{-1}	String decay rate ($\tau \approx 55$ yr)
t_n	$t - t_0 < 0$ for TRZs	s or days	Negative time
$\hat{\phi}_j$	≈ 1	–	String unit vector
$g_{\mu\nu}$	$\text{diag}(1, -1, -1, -1)$	–	Minkowski metric
η	1×10^{-22}	–	Aether coupling
$T_s^{\mu\nu}$	$\approx 1.123 \times 10^7$ J/m	J/m	Stress-energy tensor
δ_i	1.0	–	Inertial coupling
U_i	$\lambda_i \rho_{vac,[SCm]} \rho_{vac,[UA]} \omega_s(t) \cos(\pi t_n)(1 + f_{TRZ})$	J/m	Universal inertia
D_E	$\propto E^{0.5}$ (Kolmogorov)	–	CRP diffusion coefficient
$n(p)$	$p^{-2.2} e^{-p/p_{max}}$	m? (GeV/c)?	CRP momentum distribution

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3. System 2: UQFF Resonant (Oscillatory Terms)

Physical basis: Models oscillating vacuum fields via $\cos(\pi t_n)$ driving TRZs and negentropy.

Core resonant components:

$$\text{Resonant modulator: } \cos(\pi t_n) = \cos(\pi(t - t_0))$$

$$\text{String buildup: } (1 - e^{-\gamma t \cos(\pi t_n)})$$

$$\text{Reactivity decay: } e^{-\kappa t} \text{ with } \kappa = 0.0005 \text{ day}^{-1}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$\cos(\pi t_n)$	$\cos(\pi(t - t_0))$	–	Periodic reversal oscillator, period = 2 days
t_0	Initial epoch	days	Reference time
t_n	$t - t_0 < 0$	days	Negative time in TRZs
γ	0.00005 day^{-1}	day^{-1}	Decay rate, $\tau = 1/\gamma \approx 54.8$ yr
κ	0.0005 day^{-1}	day^{-1}	Reactivity decay, $\tau_\kappa \approx 5.48$ yr
$(1 - e^{-\gamma t \cos(\pi t_n)})$	$\approx \gamma t \cos(\pi t_n)$ for small t	–	Resonant buildup in U_m
f_{TRZ}	0.1	–	TRZ factor scaling U_i for negentropy
ω	π rad/s (rad/day)	rad/day	Cycle constant

Application: EP-09 (3C 273 jet ratio $R = |\cos(\pi t_{n1})/\cos(\pi t_{n2})| > 100$ for $\Delta t \approx 0.5$ days).

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4. System 3: UQFF Buoyancy (Ub_i Opposition System)

Long Form:

$$U_{b,i} = -\beta_i U_{g,i} \cdot \frac{\omega_g M_{bh}}{d_g} \cdot (1 + \delta_{sw} \lambda_{vac,sw}) \cdot [UA] \cdot \cos(\pi t_n)$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$U_{b,i}$	Full expression above	J/m	Buoyancy density (opposes $U_{g,i}$)
β_i	0.61	–	Buoyancy coupling (uniform $i=1$4)
$U_{g,i}$	See System 1	J/m	Gravity component
ω_g	7.3×10^{-16} rad/s	rad/s	Galactic rotation rate
M_{bh}	8.15×10^{36} kg	kg	SMBH mass (Sgr A*)
d_g	2.55×10^{20} m	m	Galactic distance
δ_{sw}	0.01	–	Solar wind modulation factor
$\lambda_{vac,sw}$	$\rho_{sw} c^2 \approx 7.2 \times 10^{-4}$ J/m	J/m	Wind vacuum density

Symbol	Value / Equation	Units	Description
$[UA]$	10^{-11} C	C	Trapped Aether charge
$\cos(\pi t_n)$	Oscillator	–	Time-reversal modulation

Physical interpretation: $U_{b,i}$ feeds outflows by opposing gravity. For BNS mergers (EP-11, GW170817): dynamical ejecta fraction $\approx 1 - \beta_i \approx 0.39 \approx 40\%$. For jets: $\cos(\pi t_n)$ reversals create $R > 100 : 1$ asymmetry (EP-09).

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5. System 4: UQFF Superconductive ([SCm] Reactivity)

Long Form (dual form):

$$E_{react} = 10^{46} e^{-0.0005t} \equiv \frac{\rho_{vac,[SCm]} v_{SCm}^2}{\rho_{vac,A}} e^{-\kappa t}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
E_{react}	$10^{46} e^{-\kappa t}$	W/m	[SCm] superconducting reactivity
$\rho_{vac,[SCm]}$	7.09×10^{-37} J/m	J/m	SCm vacuum energy density
v_{SCm}	1×10^8 m/s ($= c/3$)	m/s	SCm propagation velocity
$\rho_{vac,A}$	1×10^{-23} J/m	J/m	Aether vacuum density
κ	0.0005 day^{-1}	day^{-1}	Reactivity decay constant
10^{46}	$= \rho_{vac,[SCm]} v_{SCm}^2 / \rho_{vac,A}$	W/m	Quasar-scale base efficiency
$[SCm]$	10^{15}	m?	Superconductive material density
$[UA]$	10^{-11}	C	Universal Aether charge

Calibration: EP-05 (Fermi LAT 4LAC blazars): $\bar{\kappa} = 0.000497 \pm 5\%$ day^{-1} across 3,743 sources. EP-07 (Parker Solar Probe): $\delta_{sw} = 0.01 = [SSq]/57$.

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6. System 5: UQFF Triadic (Triadic Master Equations)

Long Form (e.g., Westerlund 2 plasma turbulence):

$$F_{U,\text{Triadic}} = F_U + \left(U_{g3} \cdot U_{b,i} \cdot U_m\right)^{1/3} \cdot \exp\left(-[SSq] \cdot \frac{n}{26}\right)$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$F_{U,\text{Triadic}}$	$F_U + $ geometric mean term	J/m	Triadic unified field
$(U_{g3} \cdot U_{b,i} \cdot U_m)^{1/3}$	Geometric mean of disk gravity, buoyancy, magnetism	J/m	Triadic coupling
$[SSq]$	$\log_{10}(\rho_{vac}/\lambda_{vac}) \approx 38$ (for 10^{-38} ratios)	–	Self-similar quotient
n	126 (level index)	–	UQFF ladder level
$n = 13$	Plasma systems (Westerlund 2, Pillars)	–	Triadic plasma level
$\exp(-[SSq] \cdot n/26)$	$\exp(-38 \cdot n/26)$	–	Level-scaled suppression
$Q_{wave,47}$	Mean: 3.97×10^4 J/m, std: 5.11×10^4 J/m	J/m	47-system wave energy density
Jarque-Bera	8.78 ($p = 0.012$)	–	Non-normality statistic for Q_wave
Leptokurtosis	0.037	–	Excess kurtosis of Q_wave distribution

Systems using Triadic: Westerlund 2 (turbulence), Pillars of Creation (buoyancy), TOI 1227 b (mass loss $\approx 10^{12}$ g/s), PLCK G287.0+32.9 (double relics), ASKAP J1832-0911, PSZ2 G181.06+48.47, AT2024tvd, G359.13142-0.20005.

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7. System 6: UQFF Quadratic (Quadratic Field Approximations)

Long Form (low-order approximation of potential):

$$V(r) \approx a_0 + a_1 r + a_2 r^2$$

$$T_s^{\mu\nu} = 1.27 \times 10^3 + 1.11 \times 10^7 \text{ J/m}^3 \text{ (quadratic sum)}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$V(r)$	$\sum_n a_n r^n \approx$ quadratic for low deg	J or m/s	Field potential approximation
a_0	Constant term	–	Polynomial fit coefficient 0
a_1	Linear coefficient	–	Polynomial fit coefficient 1
a_2	Quadratic coefficient ($\approx 10^{-12}$ for $n = 8$)	–	Polynomial fit coefficient 2
R^2	≈ 0.95 (for ENSDF $n = 8$ bindings)	–	Quadratic fit quality
$T_s^{\mu\nu}$	$1.27 \times 10^3 + 1.11 \times 10^7$	J/m	Stress tensor quadratic sum
$a_2^{(Pb-206)}$	$\approx 10^{-12}$	–	Fit for EP-04 Pb-206 $n = 8$ levels

Application: Nuclear binding energy polynomial fits (EP-04 ENSDF Pb-206, EP-02 PDG 2025) use degree-8 polynomials that reduce to effective quadratic forms for low-n regimes.

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8. System 7: UQFF Master Buoyancy (Extended Ub_i with Mayan Alignment)

Long Form:

$$U_{b,\text{Master}} = -\beta_i U_{g,i} \frac{\omega_g M_{bh}}{d_g} (1 + \delta_{sw} \lambda_{vac,sw}) [UA] \cos(\pi t_n) + \exp(-(p-t)) \cdot \frac{U_m}{\rho_{vac,[UA]}}$$

Physical basis: Extends standard $U_{b,i}$ with a time-scaled magnetic feed term. The `exp(-(p-t))` factor aligns with Mayan calendar time constants (Baktun cycle) in the UQFF cosmological scaling.

Variable Equations:

Symbol	Value / Equation	Units	Description
$U_{b,\text{Master}}$	Full expression above	J/m	Master buoyancy extended form
Standard $U_{b,i}$	System 3 above	J/m	Base buoyancy term
$\exp(-(\pi - t))$	For $t < \pi$ days, decays toward Euler	–	Mayan/Master time alignment factor
U_m	$\sum_j [\mu_j / r_j (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j] P_{SCm} E_{react}$	J/m	Universal magnetism (lossless strings)
$\rho_{vac,[UA]}$	7.09×10^{-36} J/m	J/m	UA vacuum energy density
P_{SCm}	≈ 1 (Sun)	–	SCm core penetration factor
E_{react}	$10^{46} e^{-\kappa t}$	W/m	Reactor efficiency

Distinction from PAPER_064: PAPER_064 documents the four simplified operational modes. This Master Buoyancy form adds the resonant magnetic feed term $e^{-(p-t)} U_m / \rho_{vac,[UA]}$, which drives enhanced negentropy in long-period TRZ cycles spanning Mayan-scale time constants (Baktun = 394 yr $1/\kappa^{0.33}$).

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9. Shared Constants Across All Seven Systems

Constant	Value	Physical Meaning
κ	0.0005 day^{-1}	E_react decay rate (EP-05 Fermi LAT)
$[SSq]$	0.57	Self-similar quotient (EP-04 Pb-206 S_n)
β_i	0.61	Buoyancy coupling (EP-10 IceCube, EP-11 GW170817)
F_{rel}	4.31×10^{33} N	Relativistic unified force scale
k_η	10^{-113}	LENR exponential damping factor
ω	π rad/day	UQFF cycle constant

Constant	Value	Physical Meaning
α	0.001 day^{-1}	Dipole/BH time decay in U_{g1}, U_{g4}
δ_{sw}	$0.01 = [SSq]/57$	Solar wind modulation (EP-07 PSP)
v_{sw}	5×10^5 m/s	Solar wind velocity (EP-07)
H_{SCm}	≈ 1	Heliosphere SCm factor
$f_{feedback}$	0.1	AGN/BH feedback factor
f_{TRZ}	0.1	TRZ negentropy factor
λ_i	1.0	Inertial coupling for U_i
p_{max}	10^{16} eV	CRP momentum cutoff (EP-10 IceCube)
γ_{CRP}	0.00005 day^{-1}	CRP decay rate ($\tau \approx 55$ yr)

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10. Cross-Reference to Implemented Code

System	Source Code	Class/Function	PAPER_064 equiv.
Compressed	`source2.cpp` L1960, `add_uqff_to_8_models.py` L24	`UQFF_Compressed`	Mode 1 ?
Resonant	`source2.cpp` L1961, `add_uqff_methods.py`	`UQFF_Resonant`	Mode 2 ?
Buoyancy	`add_uqff_to_8_models.py` L68	`F_U_Bi_i` / `UQFFMasterBuoyant`	Mode 3 ?
Superconductive	`add_uqff_to_8_models.py` L48	`UQFF_Superconductive`	Mode 4 ?
Triadic	`add_uqff_to_8_models.py` L76, `add_uqff_methods.py` L226	`UQFF_Triadic`	(new)
Quadratic	`add_uqff_to_8_models.py` L90, `add_uqff_methods.py` L291	`UQFF_Quadratic`	(new)
Master Buoyancy	`add_uqff_to_8_models.py` L68	`UQFFMasterBuoyant`	(new)

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11. Summary Table

System	Core Equation	Key Parameter	Primary Verification
Compressed	$F_U = \Sigma k_i U_{gi} - \beta_i U_{gi} \omega_g M_{bh}/d_g E_{react}$	$\kappa = 0.0005$ day^{-1}	EP-05 Fermi LAT
Resonant	$\cos(\pi t_n)$, $(1 - e^{-\gamma t \cos(\pi t_n)})$	Period = 2 days	EP-09 3C 273
Buoyancy	$U_{b,i} = -\beta_i U_{g,i} \omega_g M_{bh}/d_g (1 + \delta_{sw} \lambda_{vac,sw})[UA] \cos(\pi t_n)$	$\beta_i = 0.61$	EP-11 GW170817

System	Core Equation	Key Parameter	Primary Verification
Superconductive	$E_{react} = 10^{46} e^{-\kappa t}$	$\kappa = 0.0005$ day^{-1}, $v_{SCm} = c/3$	EP-08 JCAP
Triadic	$F_{U,tri} = F_U + (U_{g3} U_{b,i} U_m)^{1/3} e^{-[SSq] n/26}$	$[SSq] = 0.57$, $n = 13$	Q_wave_47 mean
Quadratic	$V(r) \approx a_0 + a_1 r + a_2 r^2$	$R^2 \approx 0.95$	EP-04 ENSDF Pb-206
Master Buoyancy	$U_{b,Master} = U_{b,i} + e^{-(\pi - t)} U_m / \rho_{vac,[UA]}$	π-alignment	Full TRZ cycles

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References

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS subratio is 0.103 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP) The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 113, \quad n_{\text{channel}} = 16/26$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 113$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed mat-

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is $\mathbf{10^4}$ **yr** (spin-down equilibrium): $\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m / M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis. ### §B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.103	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 113$	✓ Res-